

ORIGINAL ARTICLE

Prediction of 10-year cardiovascular disease risk by diabetes status and lipoprotein-a levels; the HellenicSCORE II+



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ABSTRACT

BACKGROUND The aim of this study was to develop an updated model to predict 10-year cardiovascular disease (CVD) risk for Greek adults, i.e., the HellenicSCORE II+, based on smoking, systolic blood pressure (SBP), total and high-density-lipoprotein (HDL) cholesterol levels, and stratified by age group, sex, history of diabetes, and lipoprotein (Lp)-a levels.

METHODS Individual CVD risk scores were calculated through logit-function models using the beta coefficients derived from SCORE2. The Attica study data were used for the calibration (3,042 participants, aged 45 (14) years; 49.1% men). Discrimination ability of the HellenicSCORE II+ was assessed using C-index (range 0-1), adjusted for competing risks.

RESULTS The mean HellenicSCORE II+ score was 6.3% (95% confidence interval (CI) 5.9% to 6.6%) for men and 3.7% (95% CI 3.5% to 4.0%) for women ($p < 0.001$), and were higher compared to the relevant SCORE2; 23.5% of men were classified as low risk, 40.2% as moderate, and 36.3% as high risk, whereas the corresponding percentages for women were 56.2%, 18.6%, and 25.2%. C-statistic index was 0.88 for women and 0.79 for men when the HellenicSCORE II+ was applied to the Attica study data, suggesting very good accuracy. Stratified analysis by Lp(a) levels led to a 4% improvement in correct classification among participants with high Lp(a).

CONCLUSION HellenicSCORE II+ values were higher than SCORE2, confirming that the Greek population is at moderate-to-high CVD risk. Stratification by Lp(a) levels may assist in better identifying individuals at high CVD risk. (Hellenic Journal of Cardiology 2024;79:3-14) © 2023 Hellenic Society of Cardiology. Publishing services by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. INTRODUCTION

Although a decline in cardiovascular disease (CVD) mortality rates has been observed over the past decade, the absolute number of individuals living with CVDs still increases, making this chronic condition the main cause of death and disability in most countries around the world¹⁻⁵. For a better

identification of the potential CVD candidate and to make effective, personalized interventions, several individual risk prediction models have been proposed through the years and used in guidelines, such as the Framingham Coronary Risk Chart in the US, the QRISK in England and Wales, and the European Society of Cardiology (ESC) Systematic Coronary Risk Estimation (ESC SCORE) for European populations,

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with its latest version (i.e., SCORE2) published in 2021, as well as many others with less widespread use⁶⁻¹⁰. Regarding the recent SCORE2, four CVD risk charts for fatal and non-fatal events were developed to reflect the large variation in incidence across European countries, using data from 45 cohort studies and 13 European countries (Greece was not included).⁹

However, it has been strongly recommended that CVD risk scores, to be adapted, should use country-specific data^{6,11-13}. The ESC SCORE and SCORE2 did not use data from the Greek population. Therefore, initially in 2007 (using baseline data from the Attica study collected between 2001 and 2002) and then in 2020 (using data from the Hellenic National Health and Nutrition Survey collected during 2015-2016), the HellenicSCORE and the recalibrated HellenicSCORE II were developed, respectively, based on the methodology of ESC SCORE but using data from the Greek population^{14,15}. Age, sex, smoking, systolic blood pressure (SBP), and total cholesterol levels were used to estimate CVD risk in HellenicSCORE and HellenicSCORE II. At this point, it should be noted that the recent SCORE2 uses non-high-density lipoprotein (non-HDL) cholesterol instead of total cholesterol and incorporates diabetes history for more accurate CVD risk estimation⁹.

Despite the significant advantages and improvements made in SCORE2 compared to the previous version, a considerable proportion of individuals remained misclassified in external validation cohorts⁹. Therefore, additional markers to improve the goodness-of-fit of the risk models should be considered. A marker that has emerged from pathophysiological, epidemiological, and genetic studies and seems to play a role in the causality of CVD is lipoprotein (Lp)a. Current evidence suggests that Lp(a) is strongly associated with CVD risk and is attributed to 1-out-of-3 CVD cases globally¹⁶⁻²⁰. Despite its difficulties in measurement, epidemiological, genome-wide association studies, and Mendelian randomization data suggest a causal role for elevated Lp(a) in the development of CVD²¹. However, controversy remains concerning the nature of Lp(a) risk curve and the extent to which it confers a better prediction of CVD risk, independently of other lipid markers used in current risk equations.

CVD risk estimation charts have been widely adopted by many cardiovascular societies around the world as an important tool for primary prevention. However, they need to regularly be updated with the latest population-based data on risk factor levels and survival rates. Thus, the aim of the present study was to develop an updated model, the HellenicSCORE II+,

to predict 10-year fatal and non-fatal CVD risk for Greek adults aged 40-69 years based on smoking, SBP, total and HDL-cholesterol levels, stratified by age group, sex, history of diabetes, and Lp(a) levels. Data from the Attica cohort study participants (2002-2022) were used for the recalibration²³.

2. METHODS

2.1. DEVELOPMENT OF HellenicSCORE II+. It has been suggested that when there is a systematic overestimation or underestimation of risk, a recalibration process is required to transport a prediction function from one setting to another²⁴. For the HellenicSCORE II+ development, we used the regression-calibration method recommended by D'Agostino et al.²⁴ Note that recalibration does not affect the effect size measures comparisons or discrimination evaluations. In the present study, values for 10-year morbidity and mortality CVD risk (probability) were calculated based on the beta coefficients derived from the SCORE2 model (using the b-coefficients for moderate-risk European cohorts) and the data from the Attica study sample for the recalibration (see below for details). In general, the recalibration formula of a risk equation is given by the following:

$$R(X, t) = 1 - S(X, t) \\ = 1 - S_0 \times e^{\sum_{i=1}^k b_i \times (X_i - \bar{X}_i)}$$

where $R(*, *)$ denotes the risk function of a vector (set) of characteristics X 's at a specific time interval t , and $S(*, *)$ denotes the survival function, which is the product of the baseline survival specific for certain regional factors (i.e., age, sex), i.e., S_0 , and a component that contains the adjustments of the beta coefficients of the original equation model, i.e., b_i . Specifically, adjustments of the b_i of the original equation model are derived by multiplying them with the difference of individual's characteristic, X_i , minus the population average of each X_i , i.e., $\bar{X}_i$. Thus, the following equation was used: Risk = $(1 - S_{\text{age, sex}} \times \exp \{b_1 * [\text{individual SBP} - (\text{pop. mean SBP})] + b_2 * [\text{individual HDL C} - (\text{pop. mean HDL-cholesterol})] + b_2 * [\text{individual total cholesterol} - (\text{pop. mean total cholesterol})] + b_3 * [\text{individual smoking} : (\text{yes:1/no:0}) - (\text{pop. prevalence of smoking})]\}$ where S is the age- and sex-specific average annual mortality and morbidity CVD rates as calculated in the Attica study²⁵.

Annual CVD rates were modeled using an age-specific Poisson model, joining the midpoints of 5-year intervals. The beta coefficients b_i , representing the $\log$ (hazard ratios), were calculated for each

characteristic (i.e., SBP, total and HDL cholesterol, and smoking) by pooling CVD mortality and morbidity results (see Appendix of SCORE2 publication, i.e.,⁹). For practical and illustration purposes, the CVD risk charts are displayed according to non-HDL cholesterol rather than total cholesterol and HDL cholesterol, as done in the SCORE2 project⁹. HellenicSCORE II+ values were then stratified by history of diabetes of individuals (yes/no) and Lp(a) levels (below 30 mg/dL, which has been suggested as the desirable and optimal test result range, between 31 and 50 mg/dL, and above 50 mg/dL)²⁰.

2.2. CALIBRATION STUDY. 2.2.1. Design. The Attica study database was used for the calibration of the HellenicSCORE II+ to the total population as well as for the calculation of 10-year CVD risk in various groups of Greek individuals. The study is a population-based cohort that was carried out during 2001–2002 and was based on a stratified random sampling from the entire Attica region in Greece (where approximately 45% of the total Greek population is living and includes Athens metropolitan city). Three follow-up waves were performed at 5-year, 10-year, and 20-year after baseline enrollment of the participants (aged 18+ years).

The study's design details, including sampling and procedures applied in all waves, have already been published^{23,25,26}.

2.2.2. Sample. The Attica study sample included 3,042 (1,514 men, mean 46 (standard deviation, SD, 1) years old and 1,528 women, mean (SD) 45 (13) years old) participants, who were free of CVD at baseline examination and had complete measurements of all required characteristics for the calibration of the HellenicSCORE II+.

2.2.3. Measurements and procedures. To calibrate the HellenicSCORE II+ as described above, population means for SBP (in mmHg), total cholesterol and HDL-cholesterol (in mg/dL) levels, as well as the prevalence of diabetes mellitus (yes/no), and smoking (current/ever, never), were retrieved from the Attica study database (combining both 2012 and 2022 waves), for men and women separately, and for specific age groups starting from 40 to 50 years, at 5-year intervals, according to the SCORE2⁹. Systolic blood pressure was measured by having participants rested and seated for at least 5 minutes; three consecutive measurements were made using validated auscultatory devices and cuffs according to the participant's arm size were used, and the mean value was calculated. Total and HDL-cholesterol concentrations were measured in mg/dL using chromatographic enzymic method in a Technicon automatic analyzer RA-1000 (Dade Behring, Marburg, Germany).

Non-HDL cholesterol was calculated using the formula: [non-HDL cholesterol] = [total cholesterol] - [HDL cholesterol], in mg/dL

Transformation of cholesterol values from mg/dL to mmol/L was made using the conversion factor 0.02586. In participants who were on lipid-lowering medications, we adjusted the lipid concentrations depending on the type of lipid-lowering medication intake based on prior reports of effect size for each medication type from the literature²¹.

Diabetes mellitus type II was defined as use of glucose control agents, insulin dependent, or fasting glucose levels above 125 mg/dL; age diabetes was diagnosed was not recorded, and there were no participants with type I diabetes in the study²³. Furthermore, Lp(a) concentration was measured in mg/dL by a latex-enhanced turbidimetric isoform-insensitive immunoassay.

In the SCORE2 equation, diabetes history was part of the risk equation, but as in our study, it was not possible to accurately assess the exact age of diabetes was diagnosed, so we used diabetes history as a stratification factor (strata).

2.2.4. Use of Risk Chart. As in the previous versions, the HellenicSCORE II+ should be used in accordance with the guidelines of the ESC SCORE chart and under the supervision of a physician⁹. We first select the chart based on specific sex and age range. Then, we follow the chart according to current smoking status and select the cell nearest to the SBP and non-HDL cholesterol levels. If the age range approaches the next age group, then adjust the risk estimate upward. As suggested by SCORE2 project investigators, the estimated CVD risk models are intended for use in people aged 40–69 years [for older ages, see SCORE-Older Persons (OP)]. The SCORE2, and consequently HellenicSCORE II+, should not be used in individuals with established CVD, chronic kidney disease, familial hypercholesterolemia, or very high levels of CVD risk factors.

2.3. STATISTICAL ANALYSIS. Stata statistical software, version 17 (STATA Corp., College Station, TX, USA) was used for all calculations. A special code was written in Stata software to calculate the HellenicSCORE II+ based on the methodology described above (it is available upon request). Stata code for calculation of the HellenicSCORE II+ algorithms is available on request. Moreover, the *score2risk* Stata program was also used to calculate the 10-year risk of CVD according to the SCORE2 algorithm, as described in the published methods paper (i.e., the SCORE2 working group and the European Society of Cardiology's Cardiovascular Risk Collaboration,⁹). Data of

TABLE 1 Summary of average levels of selected factors used for HellenicSCORE II+ risk model derivation, based on the Attica study participants

Sex	Age (yrs)	SBP (mmHg)	Total Cholesterol (mg/dL)	HDL Cholesterol (mg/dL)	Smoking prevalence (% ever)	Diabetes prevalence (%)	Lp(a) (mg/dL)
Men	40	119	181	44	61.8	0.9	14.5
	45	125	198	43	70.9	2.1	17.5
	50	131	204	42	75.9	5.6	18.4
	55	135	205	41	59.2	6.5	18.5
	60	137	205	40	55.6	16.6	18.1
	65	140	205	39	68.1	23.0	19.3
Women	40	109	170	54	50.8	1.08	16.8
	45	116	182	52	50.9	2.1	18.8
	50	124	193	51	50.3	1.9	19.7
	55	130	212	50	40.2	5.2	21.8
	60	136	216	49	31.2	12.2	21.7
	65	138	217	48	30.2	15.5	21.5

Data are presented as mean (SD) or as relative frequencies (percentages) accordingly.

continuous and categorical variables are presented as mean (standard deviation, *SD*) or as relative frequencies (percentages), accordingly. Comparisons between HellenicSCORE II+ and SCORE2 values for men and women at specific ages were made using the Wilcoxon signed ranked test for paired observations. The Spearman *rho* correlation coefficient (due to the skewed distribution of the scores) was calculated to evaluate the linear dependency between the two scores. The Bland and Altman method and the respective plot were used to evaluate concordance (or discordance) between HellenicSCORE II+ and SCORE2 values; results are presented as bias (i.e., difference between the two CVD risk scores) and the corresponding 95% confidence interval (CI). Linear regression models were applied to evaluate the association of age, sex, smoking habit, and medical history of the participants on score's bias. The accuracy of HellenicSCORE II+ and SCORE2 risk models was evaluated based on their classification ability to discriminate true CVD events during the 10-year period from CVD-free participants, using data from the Attica study. The Hosmer-Lemeshow C-index statistic of goodness-of-fit for binary outcomes (where estimated risk >0.5 is considered as an event) was used to evaluate the classification ability of HellenicSCORE II+ and SCORE2 (using *lroc* and *hl* commands). C-index ranges from 0 to 1, where values close to 1 indicate better discrimination ability.

3. RESULTS

3.1. BASELINE CHARACTERISTICS OF THE ATTICA STUDY PARTICIPANTS USED FOR THE CALIBRATION OF THE HellenicSCORE II+.

Mean values for SBP,

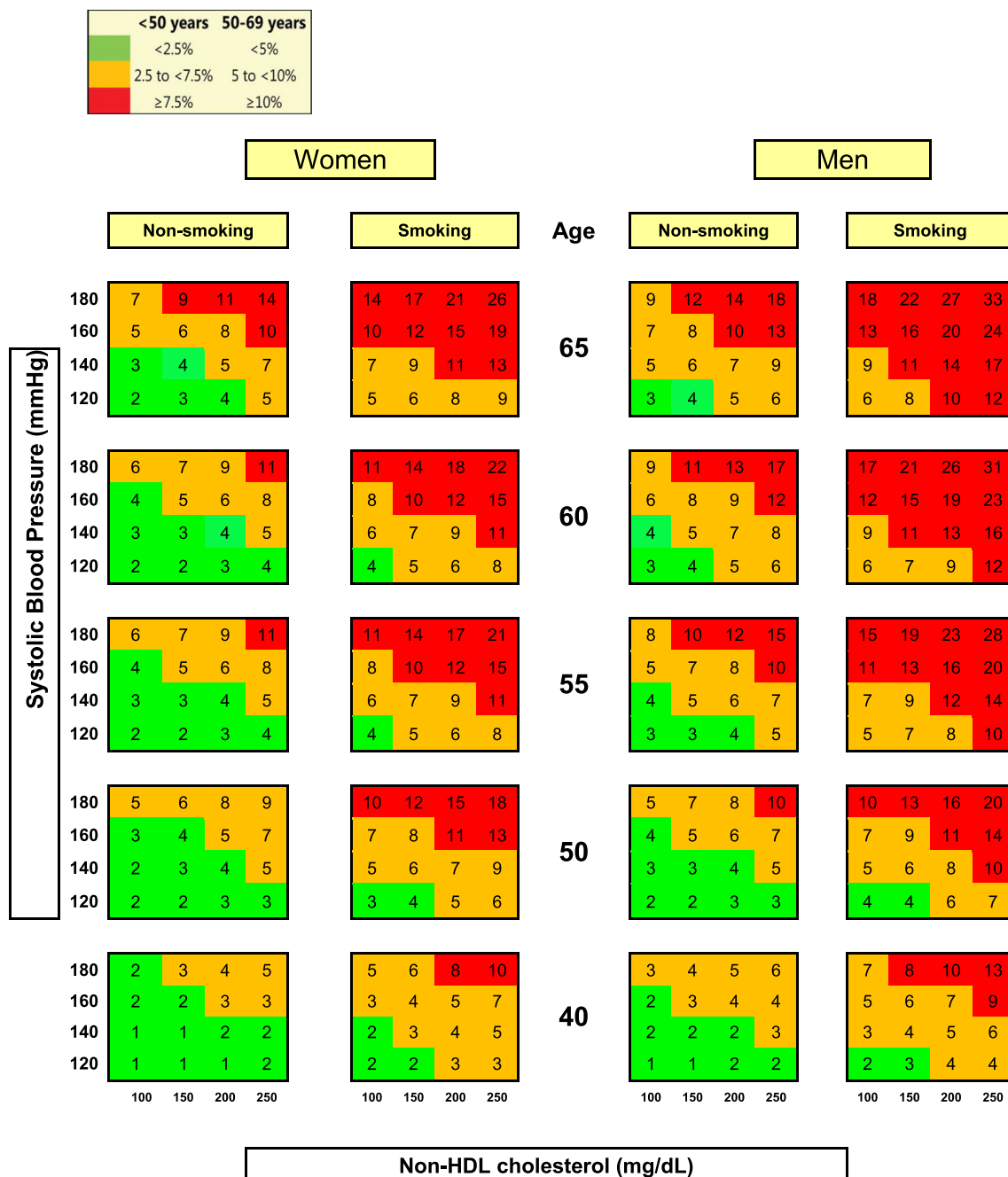
total, and HDL cholesterol, Lp(a), as well as smoking and diabetes prevalence, as derived from the available waves of the Attica study, are presented in **Table 1**, in men and women, and at various index ages. These values were used for the calibration of the HellenicSCORE II+ risk prediction tool.

3.2. TEN-YEAR RISK FOR FATAL AND NON-FATAL CVD EVENTS BASED ON THE HellenicSCORE II+.

The risk charts for 10-year combined (i.e., fatal and non-fatal) CVD events based on the recalibrated HellenicSCORE II+ are illustrated in **Fig. 1**⁹. Based on the Attica study data, the mean HellenicSCORE II+ score value of the referent general population was 6.3% (95% confidence interval (CI) 5.9% to 6.6%) for men and 3.7% (95% CI 3.5% to 4.0%) for women.

In **Table 2**, mean HellenicSCORE II+, as well as SCORE2 values, by age class and sex are presented; the values represent the estimated 10-year risk for fatal and non-fatal CVD events at mean levels of SBP, total and HDL-cholesterol levels, and adjusting for smoking status. The CVD risk varied from 0.7% to 37.1% in men and from 0.09% to 28.2% in women. A progressive increase of the predicted CVD risk was observed in both sexes, according to the aging of the population. The 10-year CVD risk was significantly higher for men compared to women at all index ages ($p < 0.001$). According to the risk classification classes suggested by SCORE2 (i.e., low: <2.5%, moderate: 2.5%-7.5%, and high: >7.5% for aged <50 and low: <5%, moderate: 5%-10%, and high: >10% for aged 50-69 years), 23.5% of men were classified as low 10-year CVD risk, 40.2% as moderate, and 36.3% as high risk, whereas the percentages for women were 56.2%, 25.2%, and 18.6% for low, moderate, and high risk, respectively.

FIGURE 1 HellenicSCORE II+ charts for estimating the 10-year risk of fatal and non-fatal cardiovascular disease events in Greece (risk estimates are linearly smoothed for better interpretation; data derived from Attica study participants)



3.3. COMPARISON BETWEEN HellenicSCORE II+ AND SCORE2. In Fig. 2 the distribution of HellenicSCORE II+ and ESC SCORE2, in men (upper plot) and women (bottom plot) is presented. The two scores showed a good linear correlation when applied to the Attica study participants ($r = 0.484$, $p < 0.001$). However, a high correlation does not necessarily

imply that there is good concordance between the two tools. Indeed, significant differences (discordance) were observed between HellenicSCORE II+ and SCORE2 values in both men and women. HellenicSCORE II+ values were on average higher by 0.53% in men (95% CI for bias 0.19%, 0.87%, $p < 0.001$) and by 0.62% in women (95% CI for bias

TABLE 2 Average 10-year risk for fatal and non-fatal cardiovascular disease events based on the HellenicSCORE II+ and the SCORE2 at selected index ages, using data from the Attica study cohort

Sex	Age	HellenicSCORE II+	SCORE2
Men	40	2.3	1.8
	45	5.0	4.5
	50	7.0	6.7
	55	9.5	8.9
	60	13.0	11.6
	65	17.6	17.1
Women	40	1.0	0.6
	45	2.2	1.8
	50	4.4	3.6
	55	7.0	5.2
	60	9.3	7.2
	65	12.7	12.3

0.39%, 0.86%, $p < 0.001$), compared to the predictions made by the SCORE2 model, and this was evident at all age classes (Table 2). Substantial differences (i.e., discordance) were observed as the

estimated CVD risk increases (upper plot, Fig. 3), whereas this discordance was more evident with progressive age, i.e., CVD risk as calculated by the HellenicSCORE II+ and the SCORE2 differed especially in older people (Fig. 3). It is notable that the estimated CVD risk values for the Attica study sample are closer to the “high” CVD risk SCORE2 chart than the “moderate” risk chart. (see9) However, in some extreme cases, i.e., advanced age (>60 years) and high SBP (i.e., >180 mmHg) and cholesterol levels (i.e., >300 mg/dL), the predicted 10-year CVD risk by the HellenicSCORE II+ may have overestimated the actual risk due to high dispersion, as there were fewer subjects and fewer events observed in such cases.

3.4. EXPLORATORY ANALYSIS. Exploratory regression analysis revealed that the discordance between HellenicSCORE II+ and SCORE2 was evident in smokers (bias -0.9%, 95% CI -1.1%, -0.7%), and in non-smokers (bias 2.6%, 95% CI 2.3%, 2.9%), in hypertensive individuals (bias -2.2%, 95% CI -2.6%, -1.8%), and in normotensive (bias 1.7%, 95% CI 1.5%, 1.9%), in diabetic (bias -5.7%, 95% CI -7.1%, -4.2%) and in non-diabetic (bias

FIGURE 2 The distribution of HellenicSCORE II+ and SCORE2, in men (upper plot) and women (bottom plot), based on the Attica study participants

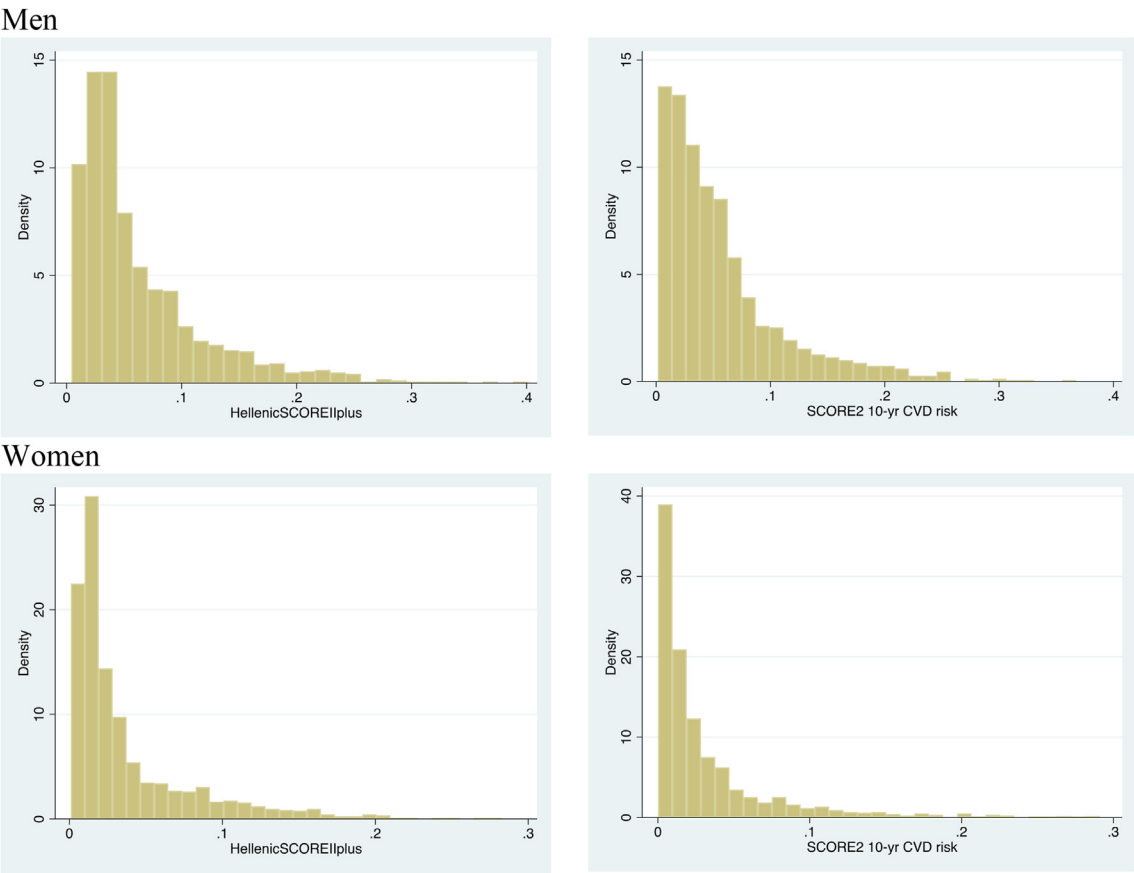
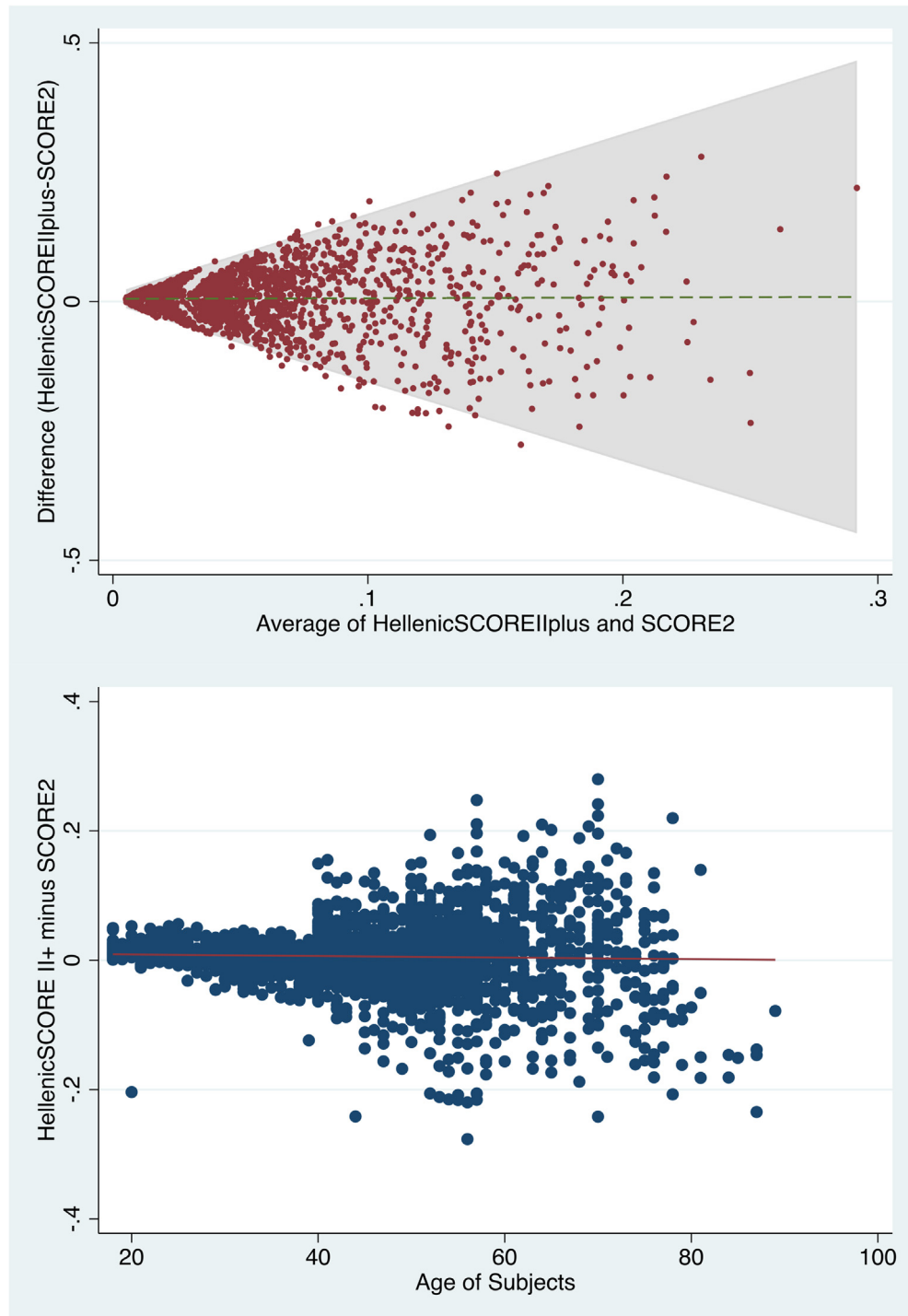
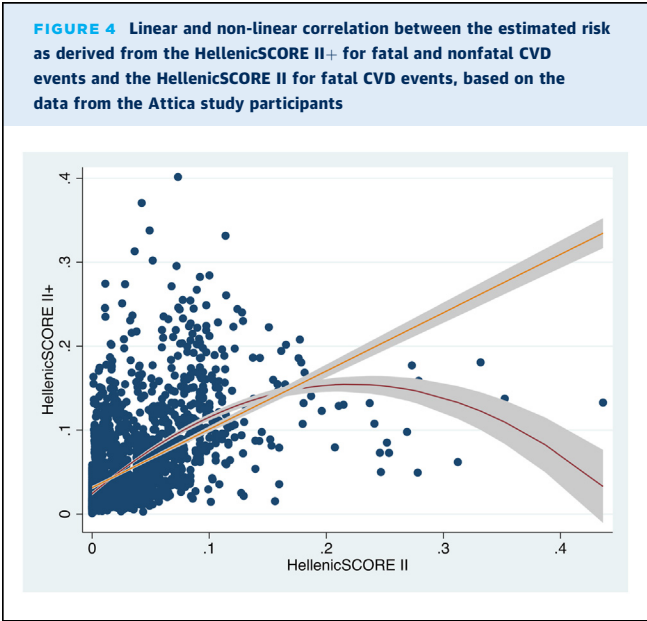


FIGURE 3 Comparisons between HellenicSCORE II+ and SCORE2 values based on the Attica study participants

1.0%, 95% CI 0.9%, 1.1%), in individuals with hypercholesterolemia (bias -0.7%, 95% CI -1.0%, -0.3%) and among those with normal lipid levels (bias 1.5%, 95% CI 1.2%, 1.7%) (note: minus sign indicates that CVD risk as estimated by

HellenicSCORE II+ was lower than the risk estimated using SCORE2).

Although direct comparison between HellenicSCORE II+ and the previous HellenicSCORE II (2021)



may be considered inappropriate because HellenicSCORE II+ estimates combined fatal and non-fatal CVD risk, whereas HellenicSCORE II estimated only fatal events, as it was found and illustrated in Fig. 4, a parabolic correlation exists, i.e., higher values of HellenicSCORE II (i.e., increased risk of fatal CVD) are associated with a modest increase in the combined fatal and non-fatal CVD risk.

Performance of HellenicSCORE II+ and SCORE2 in predicting future CVD events of the Attica study participants.

Assessment of the accuracy of HellenicSCORE II+ and SCORE2 risk tools was based on their classification ability to discriminate CVD events during a 10-year period from CVD-free participants, using data from the Attica study, and involved the calculation of

C-statistic indices. Overall, C-index revealed good discrimination, as it was 0.83 when the HellenicSCORE II+ was applied to the 10-year follow-up data of the Attica study, and it was better to the C-index when SCORE2 was considered (i.e., 0.79). Particularly, stratified by sex analysis revealed that the C-index was 0.88 for women when the HellenicSCORE II+ was applied, whereas it was 0.80 when the SCORE2 was applied, whereas for men, the C-index was found to be 0.79 and 0.77 when the HellenicSCORE II+ and the SCORE2 were applied, respectively.

Compared to the older HellenicSCORE¹⁴, or the HellenicSCORE II¹⁵, the HellenicSCORE II+ seems also to improve CVD risk discrimination (i.e., difference in C-index between HellenicSCORE II+ vs. HellenicSCORE: 0.010, $p < 0.001$; vs. HellenicSCORE II: 0.015, $p < 0.001$); this was more evident in men (i.e., difference in C-index between HellenicSCORE II+ vs. HellenicSCORE: 0.022, $p < 0.001$; vs. HellenicSCORE II: 0.025, $p < 0.001$) than in women (i.e., difference in C-index between HellenicSCORE II+ vs. HellenicSCORE: 0.009, $p < 0.001$; vs. HellenicSCORE II: 0.011, $p < 0.001$), and for the below 50 years age participants of (i.e., difference in C-index between HellenicSCORE II+ vs. HellenicSCORE: 0.030, $p < 0.001$; vs. HellenicSCORE II: 0.019, $p < 0.001$).

3.5. RISK STRATIFICATION OF CVD EVENTS BY HISTORY OF DIABETES AND LP(A) LEVELS. Men participants with type II diabetes had a 55% (at 40 years) to 190% (at 55 years) higher 10-year risk for CVD as compared to those without diabetes across index age classes, whereas for women, the risk was higher from 27% to 118% as compared to those without type II diabetes women (Table 3). The C-index was 0.82 for those without a history of diabetes and 0.84 for those with a history of diabetes. An age-adjusted

TABLE 3 Average 10-year risk for fatal and non-fatal cardiovascular disease events (adjusted for non-HDL cholesterol, systolic blood pressure, and smoking) as derived from the HellenicSCORE II+, stratified by diabetes status and lipoprotein-a levels, using data from the Attica study cohort

		Men				Women			
Age class, years		40	50	55	60	40	50	55	60
Diabetes history									
	No	1.4	5.1	7.6	9.6	0.7	3.5	6.8	7.3
	Yes	4.1	8.0	11.8	18.4	1.5	6.8	8.7	16.7
Lipoprotein-a, mg/dL									
	<30 (or 75 nmol/L)	2.2	4.9	7.2	12.0	1.0	3.4	6.1	11.7
	30-50 (or 75-125 nmol/L)	2.4	6.6	8.9	15.8	1.1	4.5	7.9	12.9
	>50 (or >125 nmol/L)	3.7	7.9	10.3	19.7	1.3	5.9	8.8	13.8

Age classes were merged as compared to the ones used in Tables 1 and 2 because of the lack of an adequate sample size in certain age groups. The conversion factor from mg/L to nmol/L used here is 2.5.

conversion factor of 1.39 for men and 1.48 for women is suggested to be applied in the derived HellenicSCORE II+ value when calculating the risk for a person with known history of type II diabetes or elevated fasting glucose levels (i.e., >125 mg/dL). For example, if for a man with known diabetes, the 10-year risk for CVD is calculated through the HellenicSCORE II+ equal to 6.5%, the diabetes-adjusted risk is $6.5\% \times 1.39 = 9.04\%$.

Regarding Lp(a) levels and 10-year CVD risk, it was observed that the inclusion of Lp(a) in the model containing HellenicSCORE II+ did not offer a significant increase in the correct classification ability of the model, i.e., C-index for the core model was 0.83, whereas it was 0.84 when Lp(a) was included as a continuous variable, which also led to substantial collinearity with the other variables in the model (p -value = 0.98). However, when the analysis was stratified by Lp(a) levels (i.e., <30 or $\geq$ 30 mg/dL), the C-index was 0.83 for those with lower Lp(a) concentrations, whereas it was 0.89 for those with higher Lp(a) levels, suggesting a 7.22% improvement of the predictive ability of the model. In addition, male participants with higher Lp(a) levels had a 30% (at 40 years) to 45% (at 65 years) higher risk for CVD as compared to those with lower levels; for women with higher Lp(a) levels, the CVD risk was higher, from 43% (at 40 years) to 77% (at 65 years) as compared to those with lower. An age-adjusted factor of 1.29 for men and 1.31 for women is proposed to be used in the HellenicSCORE II+ value when calculating the risk for a person with elevated Lp(a) levels (i.e., >30 mg/dL). Thus, for example, if for a man with Lp(a) > 30 mg/dL, the 10-year risk for CVD is calculated through the HellenicSCORE II+ equal to 4.0%, the true risk is $4.0\% \times 1.29$, which is equal to 5.16%.

4. DISCUSSION

The aim of this work was to provide an update of the HellenicSCORE II model for 10-year CVD risk prediction for Greek men and women, aged 40–69 years, in accordance with the recently presented SCORE2 model. Thus, the HellenicSCORE II+ is developed and presented here; it is a novel CVD risk score for the Greek population that incorporates both fatal and non-fatal events instead of fatal events only; and it is based on total and HDL-cholesterol levels instead of total cholesterol only, as well as on smoking, SBP levels, and is stratified by history of diabetes and Lp(a) levels.

4.1. CVD RISK PREDICTION THROUGH STATISTICAL MODELS. Understanding the distribution of CVD risk at various ages and risk factor levels is important for

the development of more effective, preventive strategies at population level and for planning a tailored intervention strategy for individuals. The development of local CVD risk models is strongly encouraged by epidemiologists around the world for accurate risk prediction and effective population healthcare planning. Aggregating countries by CVD morbidity and mortality data, even at regional level, fails to inter-grade specific cultural, environmental, genetic, and other particularities of the local reference populations into the risk models^{6,8}. In addition, it has been recommended that the CVD risk charts need to be regularly updated using contemporary data to avoid mispredictions^{11–13}. In line with these considerations, we developed the HellenicSCORE II+. It is expected that the estimation of an individual's risk for a disease would be more accurate if it is based on equations using local data from the referent population rather than global models. This new CVD risk score for the Greek population showed significant discordance with the recently presented SCORE2 (using the b-coefficients estimates for “moderate” risk countries) as compared to the HellenicSCORE II+, at all index ages. This is of importance as a considerable number of individuals would remain unaware of their risk status and consequently unmanaged through appropriate lifestyle changes and/or pharmaceutical treatment. However, this discrepancy was reduced when b-coefficients estimates for “high” risk countries were applied.

In addition, the HellenicSCORE II+ showed better performance in predicting future CVD events as compared to the older HellenicSCORE that was presented in 2007 and was based on ESC SCORE risk equations¹⁴, as well as the more recent recalibrated HellenicSCORE II that used updated prevalence data of CVD risk factor levels from a National Health and Nutrition Study (NHNS) but was also based on ESC SCORE risk estimates¹⁵. It seems that the improvements made in HellenicSCORE II+ by the incorporation of both total and HDL-cholesterol levels, as well as the stratification by diabetes mellitus status and by Lp(a) levels capture a significant proportion of the residual risk that remained unforeseen from the previous risk models. However, direct comparisons between HellenicSCORE II and HellenicSCORE II+ should be made with consideration, since HellenicSCORE II refers only to the risk of fatal CVD events, whereas HellenicSCORE II+ refers to the risk of both fatal and non-fatal events. Moreover, the correlation between HellenicSCORE II and HellenicSCORE II+ could be viewed by both a linear association, i.e., the

higher the risk for fatal CVD events, the higher the risk for both fatal and non-fatal events, as well as by a quadratic in which the risk for fatal and fatal/non-fatal CVD events increases proportionally up to the level of 20, but above that value, a decline in the risk for both fatal and non-fatal CVD events is evident (Fig. 4).

4.2. DIABETES AND ATHEROSCLEROTIC CARDIOVASCULAR DISEASE RISK PREDICTION. The latest (2021) ESC Guidelines of CVD Prevention consider type 1 diabetes, type 2 diabetes, and prediabetes as independent risk factors for CVD, which increase the risk of CVD by about two-fold, depending on the population and therapeutic control. Moreover, it is underlined that patients with diabetes are likely to have multiple CVD risk factors (i.e., obesity, dyslipidemia, and hypertension), each of which leads to an increase in the risk of CVD⁵. In line with the previous considerations, in the present study, it was found that men and women participants with diabetes had a higher risk for CVD as compared to non-diabetic. Moreover, a conversion factor was proposed here for the refinement of the HellenicSCORE II+ when calculating the risk for a person with a known history of diabetes or elevated fasting glucose levels. This conversion leads to a better understanding and estimation of the CVD risk of people with history of diabetes. However, the lack of the exact date of establishment or even diagnosis of diabetes in the Attica study may underestimate the true CVD risk.

4.3. IMPLEMENTATION OF LP(A) LEVELS IN RISK ASSESSMENT OF ATHEROSCLEROTIC CARDIOVASCULAR DISEASE. Based on a series of pathophysiological, epidemiological, and genetic studies, Lp(a) seems to play a role in the causality of CVD and explains a substantial proportion of the residual risk that low-density lipoprotein cholesterol-lowering therapies fail to capture^{16-20,27}. Epidemiological, genome-wide association studies, and Mendelian randomization data suggest a causal role for elevated Lp(a) in the development of CVD²¹. However, CVD risk estimation scores do not typically include Lp(a) as a predictor variable. In the 2022 Consensus Statement of the European Atherosclerosis Society (EAS), a new risk calculator was presented, showing how Lp(a) may influence lifetime CVD risk and how this risk may be substantially underestimated, especially in older individuals with high Lp(a) levels³⁰. A year earlier, in 2021, Patel et al., based on the data from 460,506 middle-aged UK Biobank participants, has also suggested a modified equation that could adjust the 10-year risk estimate based on Lp(a) levels²². However, it should be acknowledged that controversy remains

concerning the nature of Lp(a)-CVD risk curve and the extent to which it confers to a better prediction of CVD risk, independently of other lipid markers used in current risk equations³¹. In our study, it was observed that the inclusion of Lp(a) in the model containing HellenicSCORE II+ did not offer any significant increase in the correct classification ability of the risk model; however, when the analysis was stratified by Lp(a) levels, the correct classification rate increased by almost 4% among those with elevated Lp(a) levels (i.e., relative improvement in C-index statistic), suggesting an increase in model's true positive rate. Additionally, a conversion factor is proposed according to Lp(a) levels for the better estimation of the calculated by the HellenicSCORE II+ 10-year CVD risk.

4.3. LIMITATIONS. Risk factors' measurements from the entire area of Greece were not available, hence the calibration procedure may not be representative of the whole Greek population. It does, however, capture a large part of the population, since over 35% of the population resides in the Attica region.

The level of 100 mg/dL in Lp(a) has also been set as a high-risk threshold for CVD incidence, but we could not use it in the present modeling as there were relatively few CVD cases in the Attica study above this threshold and, therefore, led the models to be underpowered.

The exact age of diabetes mellitus type II development was not possible to determine from the Attica study participants, so diabetes was considered in the HellenicSCORE II+ as a stratification factor. In addition, during the past years, several studies have revealed that nutritional²⁸ and socioeconomic²⁹ characteristics should also be considered when assessing CVD risk to capture specific cultural and environmental particularities.

Although this was not the aim of the present work, the integration of nutritional habits and socioeconomic status may be the goal for a future project.

CONCLUSION

The updated age- and sex-specific HellenicSCORE II+ can accurately assist physicians in daily clinical practice as well as health professionals in developing effective preventive programs in the population. Based on the calculation of the HellenicSCORE II+ in the Attica study participants, the estimated 10-year CVD risk for Greek adults is at least moderate, instead of low that was considered the past decades, reinforcing the emerging need of therapeutic lifestyle changes at a population level and intensive, effective medical therapy in high-risk individuals.

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DECLARATION OF COMPETING INTEREST

None.

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REFERENCES

1. Timmis A, Vardas P, Townsend N, et al. European Society of Cardiology: Cardiovascular Disease Statistics 2021. *Eur Heart J*. 2022;43:716–799.
2. Roth GA, Mensah GA, Johnson CO, et al. Global Burden of Cardiovascular Diseases and Risk Factors, 1990–2019: Update From the GBD 2019 Study. *J Am Coll Cardiol*. 2020;76:2982–3021.
3. Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2019;139(25):e1082–e1143.
4. Michas G, Karvelas G, Trikas A. Cardiovascular disease in Greece; the latest evidence on risk factors. *Hellenic J Cardiol*. 2019;60(5):271–275.
5. Visseren FLJ, Mach F, Smulders YM, et al. ESC Scientific Document Group. 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice. *Eur J Prev Cardiol*. 2022;29(1):5–115.
6. Panagiotakos DB, Stavrinou V. Methodological issues in cardiovascular epidemiology: the risk of determining absolute risk through statistical models. *Vasc Health Risk Manag*. 2006;2(3):309–315.
7. Kavousi M, Leening MJ, Nanchen D, et al. Comparison of application of the ACC/AHA guidelines, Adult Treatment Panel III guidelines, and European Society of Cardiology guidelines for cardiovascular disease prevention in a European cohort. *JAMA*. 2014;311(14):1416–1423.
8. Damen JA, Pajouheshnia R, Heus P, et al. Performance of the Framingham risk models and pooled cohort equations for predicting 10-year risk of cardiovascular disease: a systematic review and meta-analysis. *BMC Med*. 2019;17(1):109.
9. SCORE2 working group and ESC Cardiovascular risk collaboration. SCORE2 risk prediction algorithms: new models to estimate 10-year risk of cardiovascular disease in Europe. *Eur Heart J*. 2021;42(25):2439–2454.
10. Graham IM, Di Angelantonio E, Huculeci R. European Society of Cardiology's Cardiovascular Risk Collaboration (CRC). New Way to "SCORE" Risk: Updates on the ESC Scoring System and Incorporation into ESC Cardiovascular Prevention Guidelines. *Curr Cardiol Rep*. 2022;24(11):1679–1684.
11. Menotti A, Lanti N, Puddu PE, Kromhout D. Coronary heart disease incidence in northern and southern European populations: A reanalysis of the seven countries study for a European coronary risk chart. *Heart*. 2000;84:238–244.
12. Menotti A, Puddu PE, Lanti M. Comparison of the Framingham risk function-based coronary chart with a risk function from an Italian population study. *Eur Heart J*. 2000;21:365–370.
13. Pyörälä K. Assessment of coronary heart disease risk in populations with different levels of risk. *Eur Heart J*. 2000;21:365–370.
14. Panagiotakos DB, Fitzgerald AP, Pitsavos C, Pipilis A, Graham I, Stefanadis C. Statistical modelling of 10-year fatal cardiovascular disease risk in Greece: the HellenicSCORE (a calibration of the ESC SCORE project). *Hellenic J Cardiol*. 2007;48(2):55–63.
15. Panagiotakos DB, Magriplis E, Zampelas A, Contributors; Advisory Committee. The recalibrated HellenicSCORE based on newly derived risk factors from the Hellenic National Nutrition and Health Survey (HNNHS); the HellenicSCORE II. *Hellenic J Cardiol*. 2021;62(4):285–290.
16. Waldeyer C, Makarova N, Zeller T, et al. Lipoprotein(a) and the risk of cardiovascular disease in the European population: results from the BiomarCaRE consortium. *Eur Heart J*. 2017 Aug 21;38(32):2490–2498.
17. Kouviri M, Panagiotakos DB, Chrysoshoou C, et al. Lipoprotein (a) and 10-year Cardiovascular Disease Incidence in Apparently Healthy Individuals: A Sex-based Sensitivity Analysis from ATTICA Cohort Study. *Angiology*. 2019;70(9):819–829.
18. Amiri M, Raeisi-Dehkordi H, Verkaar AJCF, et al. Circulating lipoprotein (a) and all-cause and cause-specific mortality: a systematic review and dose-response meta-analysis. *Eur J Epidemiol*. 2023;38(5):485–499.
19. Kronenberg F, Mora S, Stroes ESG. Consensus and guidelines on lipoprotein(a) - seeing the forest through the trees. *Curr Opin Lipidol*. 2022;33(6):342–352.
20. Kronenberg F, Mora S, Stroes ESG, et al. Lipoprotein(a) in atherosclerotic cardiovascular disease and aortic stenosis: a European Atherosclerosis Society consensus statement. *Eur Heart J*. 2022;43(39):3925–3946.
21. Reyes-Soffer G, Ginsberg HN, Berglund L, et al. American Heart Association Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Cardiovascular Radiology and Intervention; and Council on Peripheral Vascular Disease. Lipoprotein(a): A Genetically Determined, Causal, and Prevalent Risk Factor for Atherosclerotic Cardiovascular Disease: A Scientific Statement From the American Heart Association. *Arterioscler Thromb Vasc Biol*. 2022;42(1):e48–e60.
22. Patel AP, Wang M, Pirruccello JP, et al. Lp(a) (lipoprotein[a]) concentrations and incident atherosclerotic cardiovascular disease: new insights from a large national biobank. *Arterioscler Thromb Vasc Biol*. 2021;41:465–474.
23. Pitsavos C, Panagiotakos DB, Chrysoshoou C, Stefanadis C. Epidemiology of cardiovascular risk factors in Greece: aims, design and baseline characteristics of the ATTICA study. *BMC Publ Health*. 2003;3:32.
24. D'Agostino RB, Grundy S, Sullivan LM, Wilson P, CHD Risk Prediction Group. Validation of the Framingham coronary heart disease prediction scores: results of a multiple ethnic groups investigation. *JAMA*. 2001;286:180–187.
25. Panagiotakos DB, Georgousopoulou EN, Pitsavos C, et al. ATTICA Study group. Ten-year (2002–2012) cardiovascular disease incidence and all-cause mortality, in urban Greek population: the ATTICA Study. *Int J Cardiol*. 2015;180:178–184.
26. Damigou E, Kouviri M, Chrysoshoou C, et al. Panagiotakos DB on behalf of The ATTICA Study Group. Lifestyle Trajectories Are Associated with Incidence of Cardiovascular Disease: Highlights from the ATTICA Epidemiological Cohort Study (2002–2022). *Life*. 2023;13(5):1142, 2.
27. Gilliland TC, Liu Y, Mohebi R, et al. Lipoprotein(a), Oxidized Phospholipids, and Coronary Artery Disease Severity and Outcomes. *J Am Coll Cardiol*. 2023;81(18):1780–1792.
28. Hassen HY, Sisay BG, Van Geertruyden JP, et al. Dietary outcomes of community-based CVD

preventive interventions: a systematic review and meta-analysis. *Publ Health Nutr.* 2023 May 16: 1-12.

29. Wang F, Gao Y, Han Z, et al. A systematic review and meta-analysis of 90 cohort studies of social isolation, loneliness and mortality. *Nat Human Behav.* 2023;7(8):1307-1319.

30. Kronenberg F, Mora S, Stroes ESG, et al. Frequent questions, and responses on the 2022 lipoprotein(a) consensus statement of the European Atherosclerosis Society. *Atherosclerosis.* 2023;374:107-120.

31. Kouviri M, Panagiotakos DB. The role of lipoprotein (a) in primary and secondary

cardiovascular disease prevention: a systematic review of epidemiological studies. *Curr Opin Cardiol.* 2019;34(4):424-434.

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